MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **Weekly Test Paper 07** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

General Instructions:

- This question paper consists of **10 questions** divided into **Three Sections**: A, B, and C.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **4 questions** (Q6 of **2 marks**, Q7–Q9 of **3 marks** each).
- **Section C** comprises **1 Applied Case Study** of **4 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

Q1. What is the next term of the arithmetic progression $\sqrt{7}, \sqrt{28}, \sqrt{63}, \ldots$? [1]

(A) $\sqrt{70}$ (B) $\sqrt{84}$ (C) $\sqrt{98}$ (D) $\sqrt{112}$

Q2. Which term of the AP $121, 117, 113, \ldots$ is its first negative term? [1]

(A) 30th (B) 31st (C) 32nd (D) 33rd

Q3. If $x+1, 3x, 4x+2$ are three consecutive terms of an AP, then the value of x is: [1]

(A) 1 (B) 2 (C) 3 (D) 4

Q4. If the sum of first n terms of an AP is given by $S_n = 2n^2 + 3n$, its 5th term is: [1]

(A) 21 (B) 23 (C) 25 (D) 27

Q5. **Assertion (A):** The common difference of an AP can be positive, negative, or zero.
Reason (R): If $d = 0$, all terms of the AP are identical real constants. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.
(B) Both (A) and (R) are true, but (R) is not correct explanation.
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

SECTION B: Short Answer Questions (VSA & SA) (1 × 2 + 3 × 3 = 11 Marks)

Q6. In an AP, if the sum of its first n terms is given by $S_n = n(4n+1)$, find the general expression for its nth term, and hence determine its 10th term. [2]

Q7. If the sum of first m terms of an AP is equal to the sum of its first n terms ($m \neq n$), prove that the sum of its first $(m+n)$ terms is zero. [3]

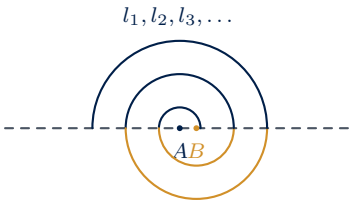
Q8. Solve the equation for x:

$$1 + 4 + 7 + 10 + \cdots + x = 287$$

[3]

Q9. If the mth term of an AP is $\frac{1}{n}$ and its nth term is $\frac{1}{m}$, prove that the sum of its first mn terms is $\frac{1}{2}(mn+1)$, where $m \neq n$. [3]

Q10. CASE STUDY: Handcrafted Circular Spiral in Textile Art



A master artisan at a handloom textile cooperative designs decorative round floor tapestries using continuous upcycled cotton yarn. The core pattern of the tapestry is woven in the form of a mathematical spiral formed by successive semicircles.

The semicircles have alternating centres at points A and B, starting with centre at A, with initial radius $r_1 = 0.5$ cm. Each subsequent semicircle increases its radius uniformly by 0.5 cm, so that the consecutive radii are 0.5 cm, 1.0 cm, 1.5 cm, 2.0 cm, $\ldots$ up to 13 consecutive semicircles.

Based on the above geometric scenario, answer the following questions (Take $\pi = \frac{22}{7}$):

(i) Write the formula for the perimeter (arc length) of a semicircle of radius r, and calculate the length of the first semicircle (l_1). **[1]**

(ii) Determine the radius and the corresponding arc length of the 13th semicircle (l_{13}). **[1]**

(iii) Find the total length of the continuous cotton yarn required to weave all 13 semicircles. **[2]**

OR

(iv) By how much does the yarn length of the 10th semicircle exceed that of the 4th semicircle? Express your answer in centimetres. **[2]**

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★★★ **END OF QUESTION PAPER · ALL THE BEST** ★★★